

Steady State Distillation

Pre-Lab Plan

Unit Operations Lab #3

Date: February 18th, 2020

Group #2 Tuesday AM section

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Commented [PTH1]: 29.5/35

1. Experimental Objective

The objective of this experiment is varying the feed position and pump speed to determine the composition at each variation and overall column efficiency using the McCabe-Thiele Method. Also, calibrate the peristaltic feed pump to create a calibration curve in order to get the actual flow rate against the position of the variable speed dial.

Commented [PTH2]: 4/5

Commented [AZ3]: Be more specific; ... in an 8-stage distillation column operating at steady state.

Commented [PTH4]: of separating ethanol from water.

2. Experimental

2.1 Apparatus

Figure 2.1 displays the continuous distillation apparatus. This apparatus includes the feed tanks where the solution will be pumped from, then goes through a heat exchanger prior entering the column. The column also includes a condenser on top of the column, and a reboiler at the bottom.

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Commented [PTH6]: missing the control computer



Figure 2.1: Label Distillation Apparatus

Table 2.1: Apparatus Descriptions

Number	Instrument	Description
1	Distillation Column	Glass distillation column with sieve trays where the VLE is obtained; Separates 10 wt% ethanol-water solution
2	Condenser	Coiled glass tubing where the cooling water flows; Cools the vapor coming out of the distillation column
3	Reboiler	Contains the feed (10 wt% ethanol) and boils feed into the distillation column
4	Heater	Used to heat the reboiler
5	Pump	Pushes feed back into reboiler
6	Funnel to Bottom	Used to charge the reboiler at the start of the experiment
7	Bottom Product Receiver	Collects the bottom product from the reboiler
8	Top Product Receiver	Collects the top product from the decanter
9	Reflux Valve	Controls the reflux within the column
10	Decanter	Decants the product coming out of the condenser into the top product tank
11	Manometer	Used to determine the pressure drop across the column
12	Control Panel	Used to manipulate the automatic valves in the process
13-20	Thermocouple 1 -Thermocouple 8	Monitors the temperature of the respective stage in the column
21	Valve 6	Located on cooling water pipe to control flow
22	Valve 7	Controls the pressure reading within the column
23	Valve 11	Controls the cooling water entering the condenser
24	Valve 5	Three-way valve that collects sample, closes reboiler, or allows for sample collection
25	Rotameter (not shown in diagram)	Used to measure the flow of cooling water through condenser
26	Temperature gauge (not shown in diagram)	Used to measure the temperature of the cooling water
27	Glass Jar	To hold the feed liquid being pumped into the column

Commented [PTH7]: the number 6 looks like it's pointing to the manometer tubing.

Commented [PTH8]: cooling water rotameter does show in the figure above.

Commented [PTH9]: "feed tank"

2.2 Materials and Supplies

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Table 2.2 Materials and Supplies

Materials/Supplies	Description
DI Water	For preparation of 10 L of mixture
Ethanol	For preparation of 10 liters of mixture
Stopwatch	Measure the time required to collect a fixed quantity and calculate flow rate
Graduated Cylinder	To fill tank and measure volume for feed pump calibration
Refractometer	Measure refractive index & composition of samples
Beakers	For sampling
Heat resistant gloves	To protect from high temperatures when handling sample
Metal Container	Acts as a sump for experiment A

Commented [PTH11]: missing funnel, used to transfer the solution into the feed tank.

2.3 Experimental Plan

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missing dependent and independent variables

Experiment A: Speed Pump Calibration

- Fill glass jar with 3000 ml of DI water.
- Disconnect the feed tube from the distillation column.
- Pump speed will be varying from 1-10 on the dial at an interval of 1.
- Allow for the flow to settle through the tube before taking a measurement.
- Measure the amount of fluid that is expelled from the tube in 10 seconds, 10 seconds is the standard time which was chosen.
- Plot the data to obtain the dial number needed to push 30 ml/min into system

Experiment B: Steady State Distillation of an ethanol-water Mixture.

- Make a 10L of 10wt% ethanol/water mixture and another solution 3L, 30wt% ethanol/water mixture. Set equipment at total reflux and ensure all valves on equipment are closed. Open V10.
- Load the 10L of ethanol through the filler cap. Fill one of the feed bottles with 3L of 30wt% ethanol/water mixture.
- Turn on the power to the control panel.
- Set the temperature selector switch to T9 and open V5 to set the cooling water flow rate to 3L/min.
- Turn the power controller for the reboiler fully counterclockwise and switch turning power to heating element to "power on" reading will be approximately **1.5kW**
- When the distillate builds up a small level in the receiver and overflows to the reflux regulator valve, Turn the reboiler power to controller to **0.75 kW**
- When the column is stable at total reflux (15-30min), the bottom product flow feed, flow of feed and reflux can be started. Set reflux ratio to 5:1.
- Open V1 bottom to let the bottom product exit and into the bottom product receiver. The Feed to the column will now go into the column through the midpoint. Set feed flow rate to 30-35 mL/min.

- After feeding 500mL, take a sample of the overhead via V3, and the bottom via V2. Take two more samples of the top and bottoms, 5 mins apart. Also record the temperatures in column T1-T9.

Experiment C: Varying the Feed Position

- Disconnect the feed tube from the column at the midpoint and reconnect the plug, plug into the top,
- When the column has reached total reflux, turn on the reboiler power to 0.75kW.
- Open valve V1 to let the bottom product exit.
- After feeding 500mL, take a sample of the overhead via V3, and the bottom via V2. Take two more samples of the top and bottoms, 5 mins apart. Also record the temperatures in column T1-T9
- Disconnect the feed tube from the column at the midpoint and reconnect the plug, plug the feed tube into the bottom. When the column has reached total reflux, turn on the reboiler power to 0.75kW.
- Open valve V1 to let the bottom product exit.
- After feeding 500mL, take a sample of the overhead via V3, and the bottom via V2. Take two more samples of the top and bottoms, 5 mins apart. Also record the temperatures in column T1-T9

• Project Safety Evaluation

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Project Title: Steady State Distillation

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	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Do not exceed required voltage, and be cautious with liquids around the electrical components
Potential mechanical hazards?	Yes	If yes, identify conditions: Distillation column, pump, potential falling hazards from heights greater than 6 feet.
Potential pressure hazards?	Yes	If yes, identify conditions: Column should be around 20 psia and potential hazard if the column is not vented to atmosphere because pressure can build up.
Potential temperature hazards?	Yes	If yes, identify conditions: The sample being taken from the bottom will be hot, temperature resistant gloves are to be worn when collecting solution.
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: Flammable liquid fatal if swallowed. If ingested or high concentration inhaled, may cause drowsiness, unconsciousness and central nervous system depression.
Gloves required?	Yes	If yes, specify type: Use <u>Latex gloves</u> while preparing and pouring the ethanol solution and heat resistant gloves when collecting samples
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water CAS #:7732-18-5, Ethanol CAS #:64-17-5
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	100C	
Pressure (psia)	20psia	Potential hazard if the column is not vented to atmosphere because pressure can build up.
Safety Controls	yes/no	If yes, Provide <u>description</u>
Safety Attire	Yes	Wear gloves, goggles, and proper clothing. Also wear heat resistant gloves when handling the <u>samples</u>

Commented [PTH14]: nitrile gloves should be good enough for preparing solution

Commented [AZ15]: How would you shut off the column in case of an emergency?

Commented [PTH16]: and fume hood for preparing ethanol solution

I have read relevant background material for the Unit Operations Laboratory entitled: “Gas Absorption in a Packed Bed” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Evelyn Almonte Feb. 18th, 2020
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